

pop_quiz projectile, vector ↕

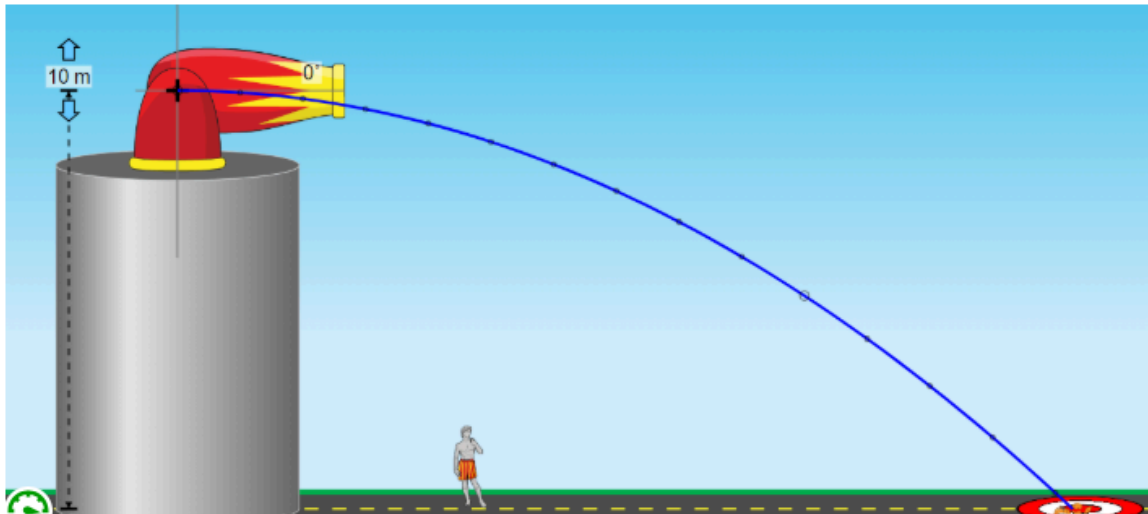
! This is a preview of the published version of the quiz

Started: Apr 29 at 7:29pm

Quiz Instructions



Question 1 25 pts



The cannon ball is fired with a speed of 1000 m/s (modern speed). The initial velocity is horizontal. You can see the height of the cannon. What is the range of the projectile ? That is the horizontal displacement (location of the target). hint: first find the time of fall .

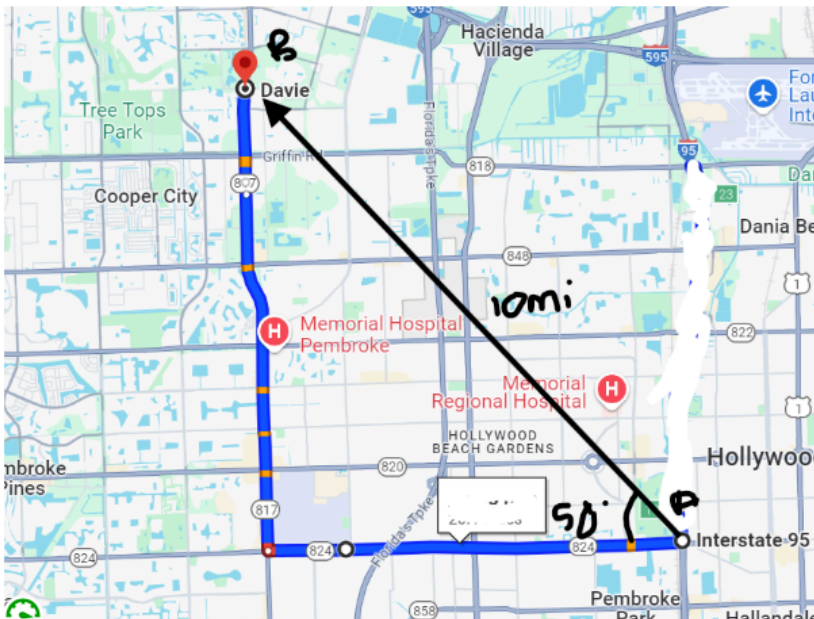


about 3200m

- ☐ about 100m
- ☐ about 2000m
- ☐ about 400m
- ☐ about 1400m



Question 2 25 pts



The vector displacement $AB = 10 \text{ miles @ } 50^\circ \text{ NW}$ (50 degrees north of west). Since you can not fly you followed the direction given by google map. What are the coordinates of the vector AB

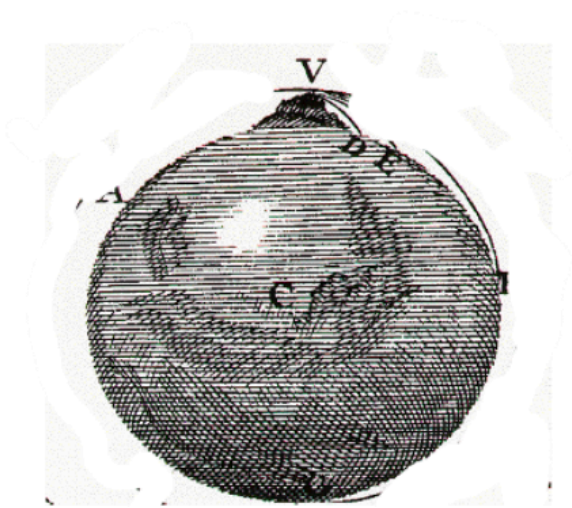
(west is -, East is +. North is +, South is -)

- ☐ $AB(-7.7, -6.4)$ miles
- ☐ $AB(6.4, 7.7)$ miles
- ☐ $AB(-6.4, 7.7)$ miles
- ☐ $AB(-7.7, 6.4)$ miles
- ☐ $AB(-10, 10)$ miles



Question 3 25 pts

2In his book principia, Newton drew a cannon fired from a high mountain. He neglected air resistance.



The trajectory of the projectile (if it lands back on earth) is:

- ☐ a parabola
- ☐ a straight line horizontal
- ☐ a straight line at an angle below the horizontal
- ☐ part of a circle



Question 4 25 pts

Newton proved mathematically that all motion in space under gravity follows a _____

- ☐ Conic section
- ☐ straight line
- ☐ spiral trajectory
- ☐ random curve

Not saved

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